Jingchang Sun

(+86)159-1093-5368 | jingchangsun@gmail.com | sun-jc.github.io

Education

Tsinghua University Sep 2016 - June 2021 (expected)
M.S. in Computer Science, Institute for Interdisciplinary Information Sciences (IIIS) Beijing

- Adviser: Prof. Pingzhong Tang, Thomas Moscibroda
- Research Area: Blockchain, Network Science, Machine Learning

Sichuan University Sep 2012 - June 2016
B.S. in Computer Science, College of Computer Science Chengdu

- GPA: 3.6/4.0 (Top 2% of Department)

Researches & Publications

Games of Miners

First author. Accepted by AAMAS'2020 (A-level-conf. in IIIS list). Patent pending.

- Studied cryptocurrency mining game;
- Derived closed-form Nash and Stackelberg equilibria;
- Showed optimal mining strategies and the effect on blockchain system security.

Nebulas Technical White Paper

Core contributor to the major technical report of a cutting-edge public blockchain. Open in July, 2017.

- Proposed and evaluated algorithms ranking blockchain addresses based on network centrality;
- Studied proof-of-address-devotion blockchain consensus

Algorithms for Computing the WCRT bound of OpenMP Task Systems with Conditional Branches

Third author. Accepted by IEEE TC (A-level-journal in IIIS list)

- Designed tools to generate task graphs for both realistic and artificial OpenMP programs;
- Analyzed response time bounds for OpenMP tasks with conditionals.

Integer Programming Approach for Schedulability of Sporadic Real-Time Systems

Second author. Published in Journal of Software (2017)

- Solved MIP formation of real-time system schedulability problem by LP relaxations;
- Implemented scheduling algorithms outperforming other state-of-the-art algorithms.

Projects & Experiences

Reliable Luxury Traceability Blockchain (patent pending) TuringSense
Nov 2019 - Sep 2019

- Developed website and smart contracts to trace and identify luxuries;
- Devised computer-vision-based approach for item identification.

Crypto-asset Graph Data Analysis Microsoft Research
Oct 2017 - Feb 2018

- Built and measured features of Ethereum user/transaction/call networks;
- Collected on-chain data and revealed behavior of token owners.

TUNet & Tsinghua-Now — Very Popular Network & Campus Helpers for Tsinghua People Tsinghua University
Nov 2016 - May 2017

- Contributed to crowdsourcing features of both Apps in both Android & iOS platform;
- Studied mobility and education data based on App Wifi usage. (summarized in a working paper)

Etherlend — A Decentralized Rotating Savings and Credit Association Tsinghua University
Nov 2016 - Dec 2016

- Leaded the project, developed smart contracts and the Android App.

Fingertip — A Heart Rate Monitor App Sichuan University
Jan 2016 - May 2016

- Leaded the development and designed signal processing algorithms.

Awards

National Scholarship & First-Class Scholarship Nov 2014
First & Second Prize of the 16th National Robot Competition July 2014

Skills

Language Proficiency: TOEFL: 105/120 | GRE: 318/340 | CET6: 555/710
Programming: Python, C, Java, Scala, MATLAB, Mathematica, R, Gurobi, Go, Solidity, Swift

孙景昶

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教育经历

清华大学 2016/09 - 2021/06 (预期毕业时间)
计算机科学与技术专业, 硕士研究生, 交叉信息研究院 (IIIS) 北京

- 导师: 唐平中 教授, Thomas Moscibroda 教授
- 研究领域: 区块链, 网络科学, 机器学习

四川大学 2012/09 - 2016/06
计算机科学与技术专业, 本科, 计算机学院 成都

- GPA: 3.6/4.0 (专业前2%)

研究与发表

Games of Miners
第一作者. 被AAMAS'2020接收 (IIIS列表A类会议). 相关专利申请中

- 研究了加密货币挖矿博弈;
- 给出并证明了闭式的Nash和Stackelberg均衡;
- 给出了最优挖矿策略以及对区块链系统安全性的影响.

星云链技术白皮书
知名公链白皮书核心作者. 全文公开于2017年7月

- 提出并评估了基于网络中心性测度的区块链地址排名算法;
- 研究了基于用户贡献度证明的共识协议.

Algorithms for Computing the WCRT bound of OpenMP Task Systems with Conditional Branches
第三作者. 被《IEEE TC》接收 (IIIS列表A类期刊)

- 设计开发了为真实OpenMP程序生成任务图的工具;
- 分析了带有条件分支的OpenMP任务集的响应时间界.

Integer Programming Approach for Schedulability of Sporadic Real-Time Systems
第二作者. 发表于《软件学报》(2017)

- 利用线性松弛求解了实时系统可调度性问题的混合整数规划模型;
- 结合Gurobi实现了更快的调度算法.

项目与经历

高可靠奢侈品溯源链 (相关专利申请中) 图灵深视
2019/11 - 2019/12

- 开发了半中心化的奢侈品溯源系统;
- 研究了基于机器视觉的实体物品身份签名算法.

数字资产图数据分析 微软研究院
2017/10 - 2018/02

- 构建以太坊用户/交易/调用网络并测量特征;
- 收集链上数据并揭示了代币持有者的行为特性.

TUNet & 即刻清华 — 清华校内流行的联网工具和教务助手 清华大学
2016/11 - 2017/05

- 贡献开发了两个应用在Android和iOS平台的数据众包功能;
- 收集应用WiFi使用信息研究了移动和教育数据 (相关内容总结于合著的工作论文中)

Etherlend — 去中心化的金融借贷应用 (开放式课程项目) 清华大学
2016/11 - 2016/12

- 带领项目开发，编写了智能合约和Android App;

十指连心 — 心率测量移动应用 (国家级大创项目) 四川大学
2016/01 - 2016/05

- 带领开发工作，设计了数字信号处理算法.

获奖情况

国家奖学金 & 校一等奖学金 2014/11
第16届全国机器人大赛一等奖 2014/07

技能

外语水平: 托福: 105/120 | GRE: 318/340 | 大学英语六级: 555/710
编程语言: Python, C, Java, Scala, MATLAB, Mathematica, R, Gurobi, Go, Solidity, Swift